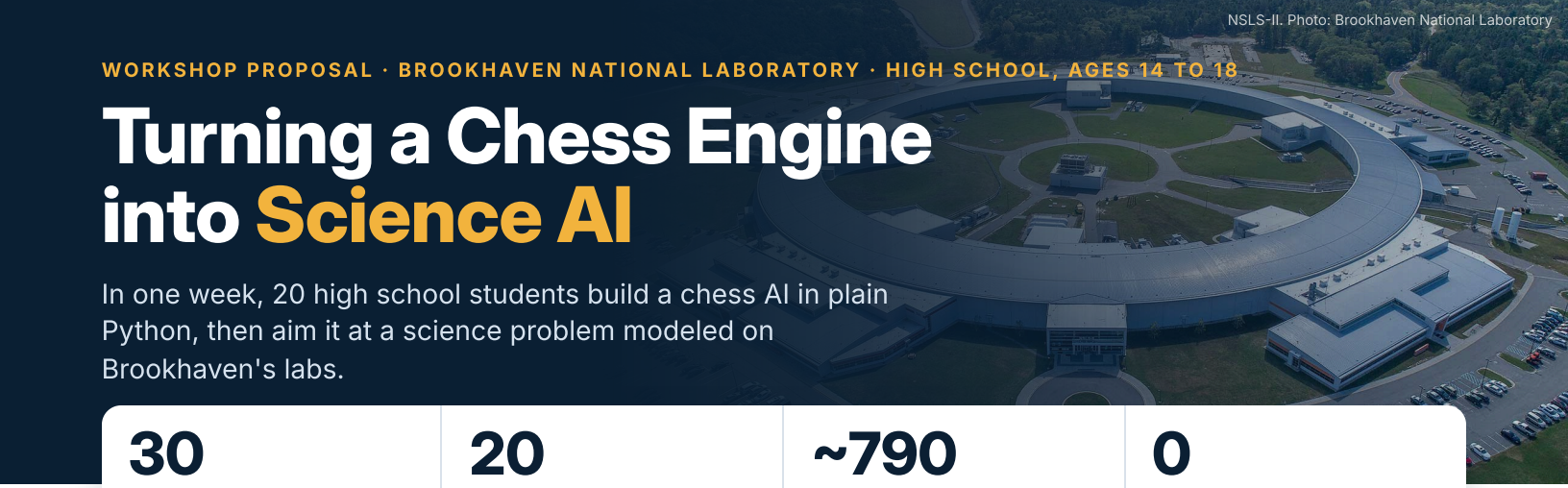


NSLS-II. Photo: Brookhaven National Laboratory

WORKSHOP PROPOSAL · BROOKHAVEN NATIONAL LABORATORY · HIGH SCHOOL, AGES 14 TO 18

Turning a Chess Engine into Science AI

In one week, 20 high school students build a chess AI in plain Python, then aim it at a science problem modeled on Brookhaven's labs.

30	20	~790	0
class hours over 5 days, four 90-minute periods a day	students in 10 student pairs, about five student pairs per adult	lines of Python each student pair types in and runs	student online accounts; everything runs offline after the Day 1 setup

Chess is where AI learned to think ahead. Science is where it is going next.

Every game-playing AI runs the same loop: describe the situation, list the possible actions, look ahead, score the results, and check the answer. An AI agent at a beamline runs that loop to choose its next measurement.

Students who already know how to code and read chess notation build **Remus**, a complete chess engine with no hidden libraries. Each day they type in the next version: Remus1 plays legal chess, Remus2 searches ahead, and Remus3 and Remus4 add the enhancements of real engines, each tested against Stockfish in Lucas Chess. Students **build a model, test it against a reference, and measure the results**.

SAME LOOP, TWO WORLDS

	Represent	Generate	Look ahead	Evaluate	Verify
IN REMUS	An 8×8 board and the full game state	Every legal move, including en passant	Minimax and alpha-beta search	Score material and position	Perft counts match a trusted reference
IN SCIENCE AI	A sample, a model, or an instrument state	Candidate measurements or hypotheses	Predict outcomes before spending beam time	Score how well each hypothesis explains the data	Compare predictions with measured data

What students will be able to do AND HOW WE WILL KNOW

OUTCOME	EVIDENCE A REVIEWER CAN CHECK	WHEN
Model a system as data	Remus1 holds every chess rule and plays only legal moves in Lucas Chess	Day 1
Read and reproduce real code	Each student pair types four engines from the instructor's code and runs each file	Days 1-4
Reason about search	Remus3 searches deeper than Remus2 in the same five seconds, shown in its info lines	Days 2-3
Measure a change	Each version plays the one before and Stockfish 1500; results compared day by day	Days 2-4
Find a hidden peak	Each student pair's AI finds a hidden hot spot in fewer readings than random guessing	Day 5
Name the crystals in a pattern	On 200 known X-ray patterns, its tree search makes fewer wrong calls than a greedy guess	Day 5
Explain a result in plain words	Each student pair presents its science results and two powders in three minutes	Day 5

The decision

1. **Approve a pilot** for 20 students, designed to grow to 30: one classroom for 5 days, 8:30 to 4:30, 10 Windows 11 laptops, and 20 USB drives.
2. **Arrange a scientist visit** on Day 5, and share one measured two-phase powder pattern from the PDF or XPD beamline for the phase hunt.
3. **Support an evaluation** with Brookhaven education staff: a concept check and an interest survey before and after the week.
4. **Judge it by evidence:** student code, games, science results, and explanations, not chess strength. No student accounts; code goes home on USB.